

Code: HUM 1207

Course Title: Bangladesh studies (History of Independence)

Marks: 60

Time: 03 Hours

1. Figures shown in the right margin indicate full marks
2. Answer Six questions, taking Three from each section including Q.1, Q.2, Q.5 and Q.6
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

- Q.1** (a) Governance means the manner in which power is exercised in the management of a country's economic and social resource for the development in an efficient and transparent way. State the significance of good governance in Bangladesh shortly. [CO2, PO7, C1] **2**
- (b) Identify in which way you can ensure good governance in Bangladesh with examples. [CO2, PO7, C2] **3**
- (c) Demonstrate 'Nine-I' theory of good governance, its application in Bangladesh, and the major challenges faced in its implementation with examples. [CO2, PO7, C3] **5**
- Q.2** The six-point movement was a Nationalist movement in the East Pakistan, which called for greater autonomy for East Pakistan. It is considered a milestone on the road to the history of the independence of Bangladesh.
- (a) Describe the cultural suppression of west Pakistan with historical examples. [CO2, PO7, C2] **2**
- (b) Sketch the significance of six-point demand in our liberation war in 1971. [CO2, PO7, C3] **3**
- (c) Interpret the historic six-point demands of 1966 and explain the perspectives that led to the Mass Uprising of 1969. [CO2, PO7, C3] **5**
- Q.3** The foreign policy of Bangladesh is shaped by its desire for peace, progress, and international cooperation. It focuses on economic growth, regional harmony, and active participation in global diplomacy.
- (a) Select the four basic principles of foreign policy of Bangladesh. [CO2, PO7, C2] **3**
- (b) Identify the key aims and objectives that shape the decisions of the foreign policy of Bangladesh. [CO2, PO7, C2] **3**
- (c) Demonstrate the foreign policy of Bangladesh adequately balance global commitments with domestic needs by referring to the principles of foreign policy for a sovereign nation. [CO2, PO7, C3] **4**
- Q.4** The Liberation War did not come overnight. It had been continuing for nine months. The war continued March to December in 1971. The sovereign and independent People's Republic of Bangladesh, as it stands today, is the outcome of an arduous struggle of the peoples of Bangladesh.
- (a) Recognize the main political parties that came together to create the United Front in the 1954 election. [CO2, PO7, C2] **3**
- (b) Discuss the major disparities between East and West Pakistan and evaluate their consequences on the eventual separation of the two regions. [CO2, PO7, C2] **3**
- (c) Demonstrate the emergence process of Bangladesh with historical examples. [CO2, PO7, C3] **4**



Section-B

- Q.5** In 1905, the British divided Bengal into two parts- one mostly Muslim and one mostly Hindu. This decision increased tensions and led to strong protests from Indian nationalists until the partition was cancelled in 1911.
- (a) State the main reason the British chose Bengal for partition. [CO1, PO1, C1] 2
 - (b) Explain the real purpose behind the British decision to divide Bengal in 1905. [CO1, PO1, C2] 4
 - (c) Describe how Indians reacted immediately after the partition and examine why their reactions were important. [CO1, PO1, C2] 4
- Q.6** After 1947, Pakistan attempted to make Urdu the only state language, ignoring the Bengali-speaking majority. Students protested, and on 21 February 1952 several were killed while demanding Bangla's recognition. Their sacrifice strengthened Bengali identity and later shaped Bangladesh's national consciousness.
- (a) List the important individuals associated with the Language Movement and mention their main roles. [CO3, PO3, C1] 2
 - (b) Describe the key factors that caused the Language Movement of 1952. [CO3, PO3, C2] 4
 - (c) Describe the key factors that caused the Language Movement of 1952. [CO3, PO3, C3] 4
- Q.7** The Agartala Conspiracy Case of 1968 accused several Bengali leaders of plotting against Pakistan. This angered the public, and by 1969, massive protests forced the government to withdraw the case. This became a key event in the struggle for autonomy.
- (a) Describe the main features of the Agartala Conspiracy Case and explain why it was historically important. [CO1, PO1, C2] 5
 - (b) Interpret the outcomes of the 1969 Mass Uprising and explain which events helped the movement grow stronger. [CO1, PO1, C3] 5
- Q.8** During the 1971 Liberation War, students, women, and freedom fighters bravely fought against oppression. Their sacrifices led to the birth of Bangladesh and continue to inspire the nation.
- (a) Explain the important roles played by students during the 1971 Liberation War. [CO1, PO1, C2] 5
 - (b) Demonstrate the areas covered by the 11 sectors of the Liberation War and name the commanders responsible for each sector. [CO1, PO1, C3] 5

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Section-A

- Q.1** (a) Define concurrent coplanar forces. [CLO1, PLO1, C1] 2
- (b) Compute the X and Y components of force P shown in **Figure: Q.1 (b)** below with respect to an (Forces can be resolved with respect to any mutually perpendicular axes oriented about any angle or inclined plane based on a given problem)
- (i) axis parallel to the horizontal plane
 - (ii) axis parallel to inclined plane. [CLO2, PLO2, C3] 4

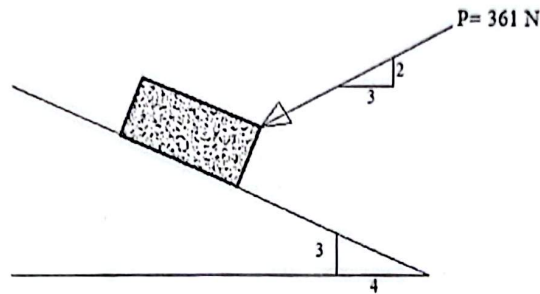


Figure: Q.1 (b)

- (c) Block P and Block Q of mass m kg is suspended through the chord which is in equilibrium position as shown in **Figure: Q.1 (c)**. Determine the mass of the Block Q. [CLO2, PLO2, C3] 4

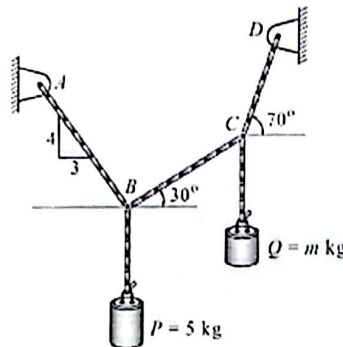


Figure: Q.1 (c)

- Q.2** (a) Define flexible chord. [CLO1, PLO1, C1] 2
- (b) A cable supported roof carries a uniform load $w = 0.6$ kip/ft as shown in **Figure: Q.2 (b)**. If the cable sag at mid-span with parabola behavior of y is set 10 ft, determine the tension in the cable (i) at points B and C and (ii) between points A and B. Neglect the weight of the cable. [CLO2, PLO2, C3] 4

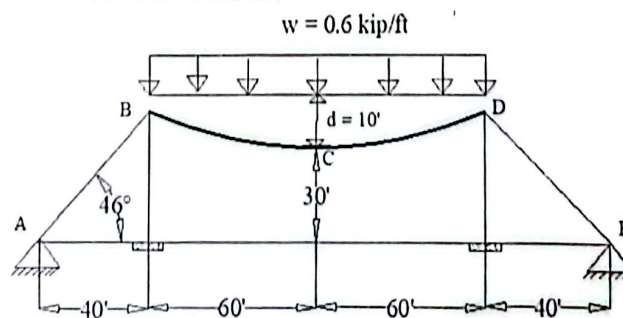


Figure: Q.2 (b)

- (c) A cable is suspended with its ends at the same elevation and 200 ft apart. The load is uniformly distributed horizontally. When the sag is 5% of the span, the maximum tension is 2040 lb. Determine the load in pounds per foot. [CLO2, PLO2, C3]

- Q.3 (a) If the resultant force is required to act along the positive u-axis and have a magnitude of 5 kN as shown in Figure: Q.3 (a), determine the required magnitude of F_B and its direction θ (the angle of the cable at the support). [CLO2, PLO2, C3]

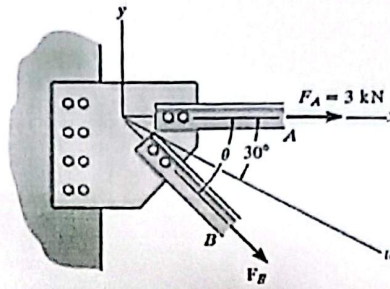


Figure: Q.3 (a)

- (b) The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon as shown in Figure: Q.3 (b), towards the other five angular points, taken in order. Determine the magnitude and direction of the resultant force. [CLO2, PLO2, C3]

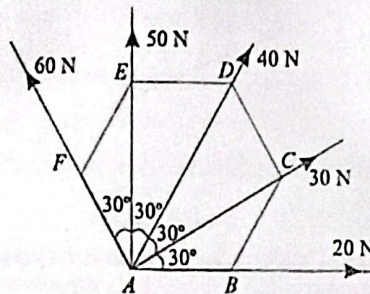


Figure: Q.3 (b)

- Q.4 (a) A block of weight 500 N is resting on the inclined plane as shown in Figure: Q.4 (a) making an angle of 33° to the horizontal. Calculate the value of force P applied parallel to the plane to just move the block up the plane if the coefficient of friction $\mu = 0.14$. [CLO2, PLO2, C3]

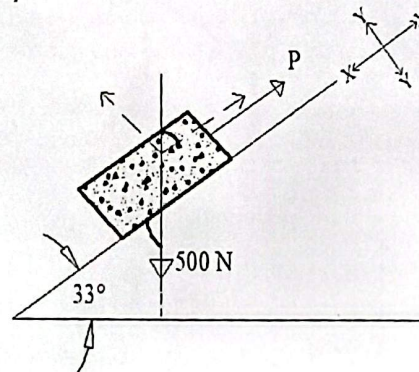


Figure: Q.4 (a)

- (b) An effort of 200 N is required just to move a certain body up an inclined plane of angle 15° [Figure: Q.4 (b)]. If the angle of inclination of the plane is made 20° as shown in Figure: Q.4 (b), the effort required is found to be 230 N. Determine the weight of the body and the coefficient of friction. [CLO2, PLO2, C3]

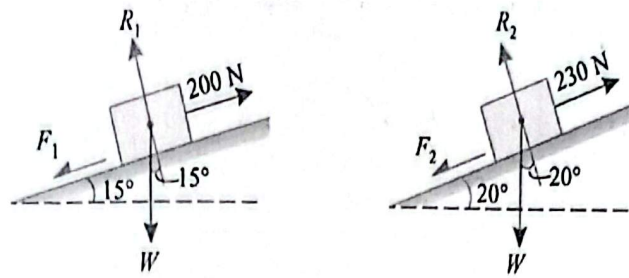


Figure: Q.4 (b)

Section-B

- Q.5 (a) State the three fundamental equations of equilibrium used in the Universal Force System. [CLO1, PLO1, C1] 2
- (b) Calculate various support reactions for the beam shown in Figure: Q.5 (b). [CLO3, PLO2, C3] 4

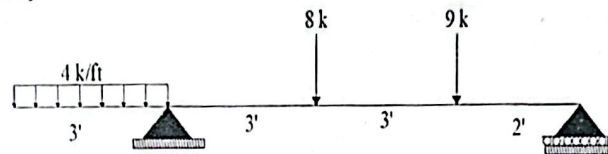


Figure: Q.5 (b)

- (c) Determine the support reactions of the following frame structure shown in Figure: Q.5 (c). [CLO3, PLO2, C3] 4

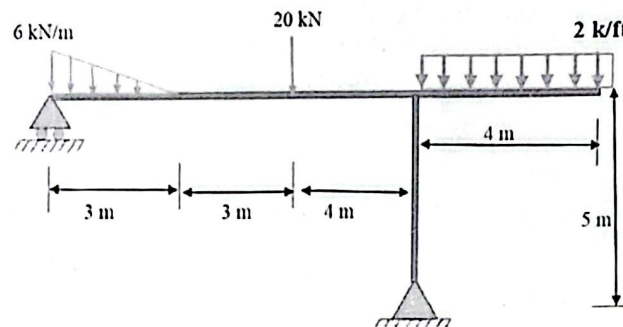


Figure: Q.5 (c)

- Q.6 (a) Define moment of inertia and radius of gyration. [CLO1, PLO1, C1] 2
- (b) Determine the moment of inertia for the shaded region shown below Figure: Q.6 (b) with respect to X and Y axes. [CLO3, PLO2, C3] 4

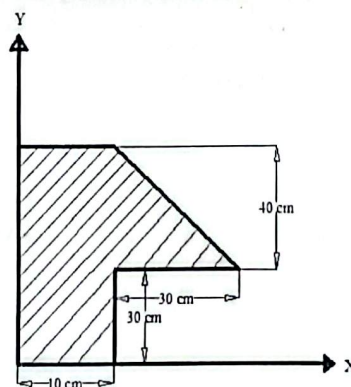


Figure: Q.6 (b)

- (c) A hollow triangular section shown in Figure: Q.6 (c) is symmetrical about its vertical axis. Compute the moment of inertia of the section about the base BC. [CLO3, PLO2, C3] 4

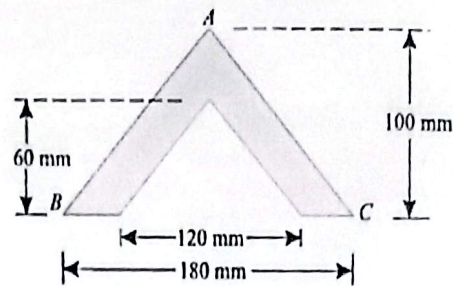


Figure: Q.6 (c)

- Q.7 (a) Determine the center of gravity of a $100 \text{ mm} \times 150 \text{ mm} \times 30 \text{ mm}$ T-section as shown in Figure: Q.7 (a). [CLO3, PLO2, C3] 5

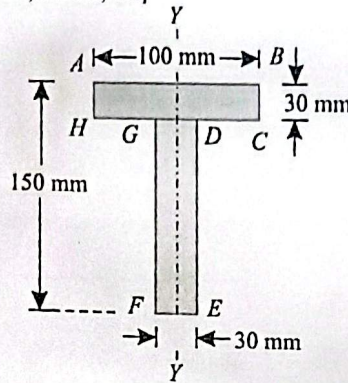


Figure: Q.7 (a)

- (b) An I-section as shown in Figure: Q.7 (b) has the following dimensions in mm units: Bottom flange = 300×100 , Top flange = 150×50 , Web = 300×50 . Determine mathematically the position of the centre of gravity of the section. [CLO3, PLO2, C3] 5

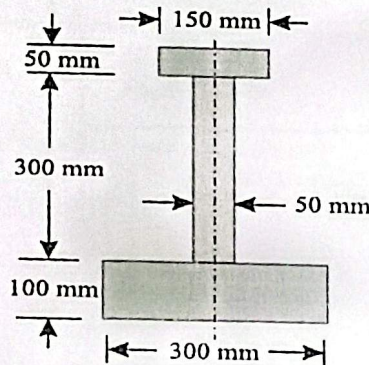


Figure: Q.7 (b)

- Q.8 (a) Determine the force in each member of the truss as shown in Figure: Q.8 (a) by the method of joints. [CLO3, PLO2, C3] 5

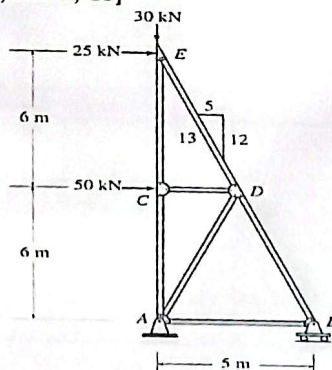


Figure: Q.8 (a)

- (b) As shown in Figure: Q.8 (b) below, in the truss, $F = 60 \text{ tons}$ and angle $\angle BDA = 25^\circ$ and $\angle BAD = 65^\circ$. Determine the external reaction components at C and the force 5

members AB and AD using the method of sections. [CLO3, PLO2, C3]

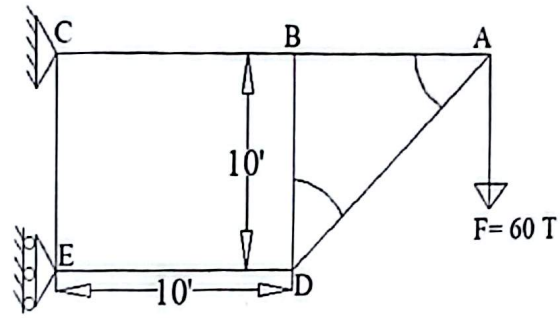


Figure: Q.8 (b)

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Section-A

Q.1 (a) Suppose, a circuit contains two loops sharing one common node.
Explain step by step how you would: [CO1, PO1]

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- Apply KCL at the common node.
- Apply KVL in each loop.
- Determine the branch currents from the resulting equations.

(b) Calculate the equivalent resistance R_{ab} in the circuit of Fig. 1(b). [CO2, PO2]

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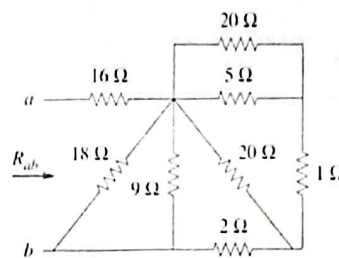


Fig. 1 (b)

(c) For the circuit shown in Fig. 1 (c), find: (i) v_1 and v_2 , (ii) the power dissipated in the 3 k Ω and 20 k Ω resistors, and (iii) the power supplied by the current source. [CO2, PO2]

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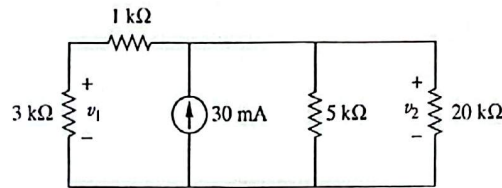


Fig. 1 (c)

Q.2 (a) Calculate the node voltages in the circuit shown in Fig. 2 (a). [CO2, PO2]

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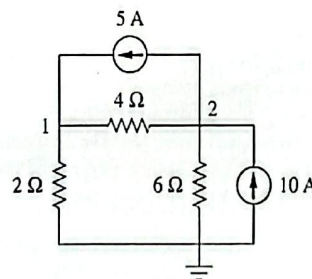


Fig. 2 (a)

(b) Determine V_0 & V_x in the following circuit shown in Fig. 2(b). [CO2, PO2]

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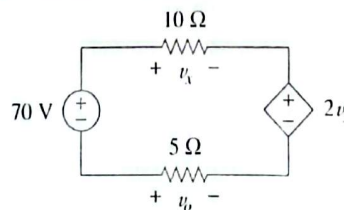


Fig. 2 (b)

- Q.3 (a) Using the superposition theorem, find v_o in the circuit of Fig. 3 (a). [CO1, PO1]

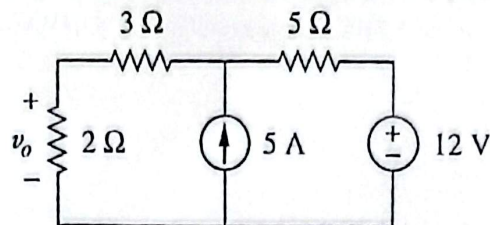


Fig. 3 (a)

- (b) Determine the value of R_L for maximum power transfer in the circuit of Fig. 3(b), and also calculate the maximum power. [CO2, PO2]

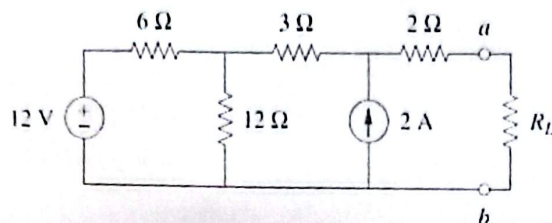


Fig. 3(b)

- Q.4 (a) Explain the I-V characteristics of a P-N junction diode with a well-labeled diagram and discuss the significance of the key points on the curve. [CO1, PO1]
 (b) Determine V_o , I_1 , I_{D1} , and I_{D2} for the parallel diode configuration of Fig. 4 (b). [CO2, PO2]

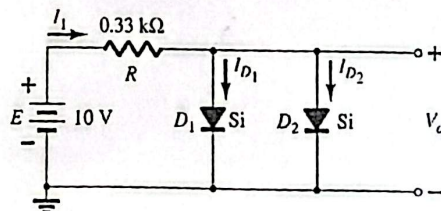


Fig. 4 (b)

Section-B

- Q.5 (a) Explain Faraday's laws of electromagnetic induction and how they govern the operation of a DC generator. [CO1, PO1] 3
 (b) In a long-shunt compound generator, the terminal voltage is 230 V when the generator delivers 150 A. Determine (i) induced e.m.f. (ii) total power generated and (iii) distribution of this power. Given that shunt field, series field, divertor and armature resistance are 92 ohm, 0.015 ohm, 0.03 ohm, and 0.032 ohm, respectively. [CO2, PO2] 3
 (c) In a no-load test of a single-phase transformer, the following data were recorded: Primary voltage: 240 V, Secondary voltage: 120 V, Primary current: 0.6 A, Power input: 40 W, and Resistance of the primary winding: 0.8 ohm. Determine the following: (i) The turns ratio of the transformer, (ii) The magnetizing component of the no-load current, (iii) The working (or loss) component of the no-load current, and (iv) The iron loss of the transformer. [CO2, PO2] 4
- Q.6 (a) In the circuit of Fig. 6(a), calculate the average power absorbed by the resistor and capacitor. Find the average power supplied by the voltage source. 5

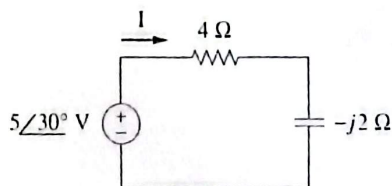


Fig. 6(a)

- Q.6 (b) The waveform shown in Fig. 6 (b) is a half-wave rectified sine wave. Find the rms value and the amount of average power dissipated in a $10\ \Omega$ resistor. [CO2, PO2] 5

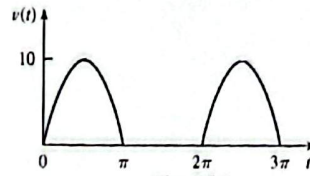


Fig. 6(b)

- Q.7 (a) Find the phase angle between: [CO2, PO2] 5
 $i_1 = -4 \sin(377t + 55^\circ)$ and $i_2 = 5 \cos(377t - 65^\circ)$
 Does i_1 lead or lag i_2 ?
 (b) An integrodifferential equation is given as -- 5

$$4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

Determine the current $i(t)$ by transforming the sinusoid into phasor. [CO2, PO2]

- Q.8 (a) Explain the purpose of rectification. Draw a center-tapped full-wave rectifier and describe its circuit operation. [CO1, PO1] 5
 (b) Determine the output waveform for the network of Fig. 8 (b) and calculate the output DC level and the required PIV of each diode. [CO2, PO2] 5

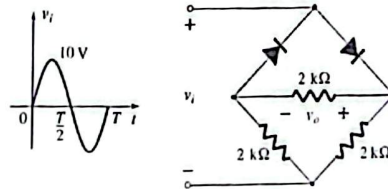


Fig. 8 (b)